Do not start attempting the test paper until you are asked to do so.

SILVER ZONE FOUNDATION
NEW DELHI, INDIA

ENROLLMENT NUMBER: (12 or 13 digit)

STUDENT NAME:

Class 6

Maximum Marks : 100
Duration : 40 Minutes
Total Questions : 50

Test Paper Type
AD

Level - 2

10th iOS INTERNATIONAL OLYMPIAD OF SCIENCE 2018-2019

Instructions for the Candidates

1. Write your 12 or 13 digit iOS Enrollment Number and your name on top of the Question Paper in the given space.
2. Filling up of improper Enrollment Number may lead to unavailability of Result.
3. This Question Paper cum Answer Sheet must be submitted to your teacher after the completion of the test.
4. Tick ☑ your answer (A, B, C, D or E) on the given box ☐ in this Question Paper cum Answer Sheet with HB Pencil or Black/Blue Ball point Pen.
5. The Question Paper cum Answer Sheet Booklet consists of 50 questions, divided into three sections.
6. Section-A: Science (30 questions); Section-B: Reasoning and Aptitude (10 questions); Section –C: Scholar's Zone (10 questions).
7. Each question of Section-A carries 1.5 marks; Section-B carries 2.5 marks and that of Scholar's Zone carries 3 marks.
8. Use of calculator or any other devices in the examination is strictly prohibited.
9. This Question Paper contains 8 pages.
10. Rough work should be done on the blank sheet/space provided in the question paper booklet.
11. Do not start attempting the test paper until you are asked to do so.
12. All the questions are compulsory and there is no negative marking.

Time Consumed ☐ Mins.

SIGN./NAME OF THE STUDENT

SIGN. OF THE TEACHER

1. Which one of the following minerals is essential for developing strong bones and teeth?
   A. ☐ Calcium and phosphorus
   B. ☐ Iodine and sodium
   C. ☐ Iron and phosphorus
   D. ☐ Sodium and iron
   E. ☐ None of these

2. Ascorbic acid is the scientific name of_____.
   A. ☐ Vitamin C B. ☐ Vitamin B
   C. ☐ Vitamin A D. ☐ Vitamin D
   E. ☐ None of these

3. _____is a water soluble vitamin whereas _____is fat soluble.
   A. ☐ Vitamin B, Vitamin C
   B. ☐ Vitamin B, Vitamin A
   C. ☐ Vitamin D, Vitamin E
   D. ☐ Vitamin A, Vitamin K
   E. ☐ None of these

4. Which minerals among the following is essential for making haemoglobin present in our red blood cells?
   A. ☐ Phosphorus B. ☐ Iodine
   C. ☐ Iron D. ☐ Potassium
   E. ☐ None of these

5. Which one of the following is not a symptom of the kwashiorkor disease in children?
   A. ☐ The stomach of child becomes swollen and bulges out.
   B. ☐ The growth of the child is stunted.
   C. ☐ The hair of child changes colour and becomes red.
   D. ☐ The ribs of child becomes very lean and thin.
   E. ☐ None of these

6. The mode of nutrition in a non-green plant is_____.
   A. ☐ saprophytic B. ☐ heterotrophic
   C. ☐ autotrophic D. ☐ holozoic
   E. ☐ None of these

7. Which of the following steps comes first in the production of woollen yarn from sheep?
   A. ☐ Scouring B. ☐ Shearing
   C. ☐ Combing D. ☐ Spinning
   E. ☐ None of these

8. Which of the following is an incorrect match?
   A. ☐ Lama — Wool yielding animal
   B. ☐ Scouring — Cleaning fleece of sheep
   C. ☐ Cocoon — Wool yielding animal
   D. ☐ Mulberry leaves — Food of silkworm
   E. ☐ None of these

9. Few steps are given below to separate salt from sand. Arrange the given steps in a correct sequence.
   i. Decantation is done.
   ii. Water is added to the mixture of sand and salt.
   iii. Stir the mixture of salt, sand and water.
   iv. Evaporation is done.
   A. ☐ i, ii, iii, iv B. ☐ iii, iv, i, ii
   C. ☐ ii, iii, i, iv D. ☐ ii, i, iv, iii
   E. ☐ None of these

10. Which one of the following is a nocturnal animal?
   A. ☐  Bat
   B. ☐  Hedgehog
   C. ☐  Chimpanzee
   D. ☐ Both A and B
   E. ☐ None of these

11. When few drops of dilute iodine solution is added to any food substance containing starch, it will turn into:
   A. ☐ dark blue B. ☐ colorless
   C. ☐ red D. ☐ violet
   E. ☐ None of these

12. A candle starts burning only when heat energy is supplied to it by the burning matchstick. The energy required to start a chemical reaction is known as its_____.
   A. ☐ activation energy B. ☐ potential energy
   C. ☐ physical energy D. ☐ mechanical energy
   E. ☐ All of these

13. Which of the following is an example of the formation of a mixture?
   A. ☐ Rust forming on an iron nail.
   B. ☐ Sugar crystals dissolving in water.
   C. ☐ Sodium and chlorine forming table salt.
   D. ☐ Hydrogen and oxygen reacting to produce water.
   E. ☐ None of these

14. The process of obtaining jute fibers from stem is called:

A. ☐ ginning B. ☐ retting
C. ☐ bobbin D. ☐ weft
E. ☐ None of these

15. The figure given below is showing which of the following kinds of separation?



A. ☐ Threshing B. ☐ Sieving
C. ☐ Filtration D. ☐ Winnowing
E. ☐ None of these

16. _____is also known as tailbone.

A. ☐ Cervical B. ☐ Coccyx
C. ☐ Sacrum D. ☐ Thoracic
E. ☐ None of these

17. Which of the following statements is correct?

I. A long flat bone located in the dorsal part of the chest is called the sternum or breastbone.

II. A tendon is a fibrous connective tissue in our body that connects a muscle to a bone.

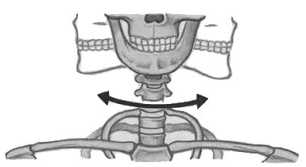
III. The pelvic girdle is a ring-like bony structure, located in the lower part of the trunk.

A. ☐ Statement I only
B. ☐ Statements I and II only
C. ☐ Statements I and III only
D. ☐ All of these
E. ☐ None of these

18. Which one of the following organs is not protected by rib cage?

A. ☐ Heart B. ☐ Lung
C. ☐ Urinary bladder D. ☐ Liver
E. ☐ None of these

19. Identify the joint given below in figure.



A. ☐ Pivot joint B. ☐ Ball and socket joint
C. ☐ Hinge joint D. ☐ Fixed joint
E. ☐ None of these

20. A force of 100 N moves a body through a distance of 4.5 m in the direction of the force. Calculate the work done.

A. ☐ 400 J B. ☐ 450 J
C. ☐ 500 J D. ☐ 475 J
E. ☐ None of these

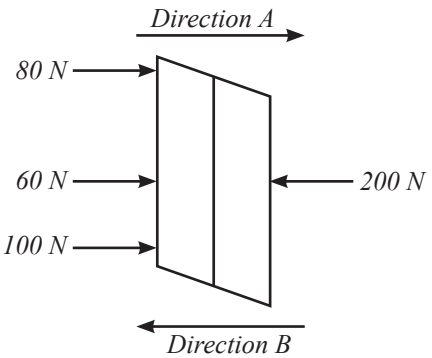
21. Which one of the following statements is incorrect?

A. ☐ Statement 1: If two forces of equal magnitude act in opposite directions on an object, the resultant force is zero.
B. ☐ Statement 2: If two forces act in the same direction on an object, the resultant force is zero.
C. ☐ Statement 3: If two forces act in opposite directions on an object, the resultant force is the difference of the two forces.
D. ☐ All of these
E. ☐ None of these

22. A train travels from Delhi with a velocity of 70km/h northwards. What is its displacement after 5 hours?

A. ☐ 150 km north B. ☐ 250 km north
C. ☐ 350 km north D. ☐ 450 km north
E. ☐ None of these

23. Calculate the resulting force and direction in which the door moves shown in the picture below.



A. ☐ Resulting force of 40 N and door moves in the direction of A.
B. ☐ Resulting force of 50 N and door moves in the direction of B.
C. ☐ Resulting force of 100 N and door moves in the direction of A.
D. ☐ Resulting force of 20 N and door moves in the direction of B.
E. ☐ None of these

24. Steve rolls a ball on smooth surface and then on rough surface. The ball moves longer distance on the smooth surface than the rough surface. Which one of the following factors is affecting the movement of the ball?

A. ☐ Force of gravity B. ☐ Force of friction
C. ☐ Force of tension D. ☐ All of these
E. ☐ None of these

25. We place an object X in between a reflecting surface and a screen Q, on which an image of the object X is formed. If a clear image of the object X is formed on the screen Q, then which of the following properties is true for the object X?

A. ☐ Transparent B. ☐ Translucent
C. ☐ Opaque D. ☐ All of these
E. ☐ None of these

26. In which one of the following cases a drought occurs?

A. ☐ When it rains heavily.
B. ☐ When it does not rain for a long period.
C. ☐ When water cycle in nature stopped working.
D. ☐ When excessive water is released in air by plants in the process of transpiration.
E. ☐ None of these 1

27. If you dump kitchen waste in a pit, it may, after sometime:

A. ☐ convert into vermicompost.
B. ☐ convert into compost.
C. ☐ remain as such.
D. ☐ remain forever in its dried form.
E. ☐ None of these

28. Our Sun is a part of which of the following galaxies?

A. ☐ Andromeda B. ☐ Orion
C. ☐ Milky way D. ☐ Comet
E. ☐ None of these

29. The acronym of BOD is:

A. ☐ biological oxygen demand
B. ☐ bacterial oxygen demand
C. ☐ biological oxide demand
D. ☐ biological ozone demand
E. ☐ None of these

30. In an ecosystem, bacterias are the:

A. ☐ producers
B. ☐ primary consumers
C. ☐ secondary consumers
D. ☐ decomposers.
E. ☐ None of these

Section–B (Reasoning and Aptitude)

31. Complete the series given below.

79, 87,____, 89, 83, 91

A. ☐ 82 B. ☐ 81
C. ☐ 48 D. ☐ 62
E. ☐ None of these

32. Pointing to a lady in a photograph Rahul said, "She is the mother of my mother's husband". How is the lady related to Rahul?

A. ☐ Mother
B. ☐ Aunt
C. ☐ Sister
D. ☐ Grandmother (paternal)
E. ☐ None of these

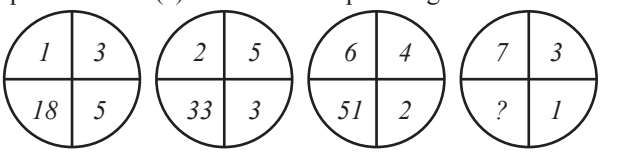
33. Mr. Das left for his office by his car. He drove 12 km towards North and then 10 km towards West. He then turned to South and covered 4 km. Further, he turned to East and moved 8 km. Finally, he turned right and drove 8 km. How far and in which direction was he from his starting point?

A. ☐ 2 km, West B. ☐ 2 km, East
C. ☐ 4 km, North D. ☐ 2 km, South
E. ☐ None of these

34. If 'ENTERTAINMENT' is coded as 'VMGVIGZRMNVMG', then 'TREATMENT' will be coded as:

A. ☐ GIZVGNMVG B. ☐ GIVZGNVMG
C. ☐ VIGZGNMVG D. ☐ GIGZGNVMV
E. ☐ None of these

35. Which one of the following numbers will replace the question mark(?) in the number pattern given below?

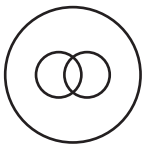


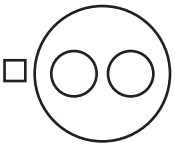
A. ☐ 24 B. ☐ 26
C. ☐ 28 D. ☐ 21
E. ☐ None of these

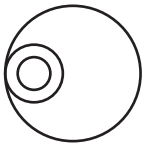
36. In a certain code language 'you are brave' is written as 'ma sa ka', and 'you and me' is written as 'ra na ma'. What is the code for 'you'?

A ☐ ma B. ☐ ka
C. ☐ sa D. ☐ ra
E. ☐ None of these

37. Which of the following figures represents the relation between 'sparrow', 'birds' and 'crows'?

A. ☐ 

B. ☐ 

C. ☐ 

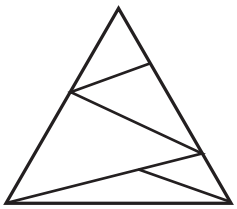
D. ☐ 

E. ☐ None of these

38. In a certain code 'GUEST' is written as '53@$2' and 'MEAN' is written as '6@4#'. How is 'SAME' written in that code?

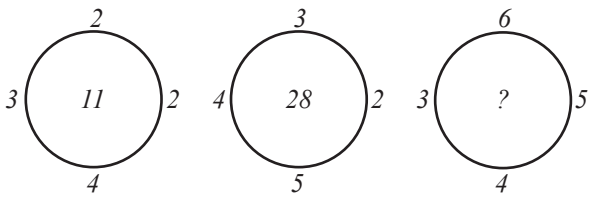
A. ☐ 4$6@
B. ☐ $46@
C. ☐ $36@
D. ☐ 5$6@
E ☐ None of these

39. Count the number of triangles in the figure given below.



A. ☐ 9
B. ☐ 10
C. ☐ 15
D. ☐ 13
E. ☐ None of these

40. Find the missing number.



A. ☐ 30
B. ☐ 36
C. ☐ 70
D. ☐ 60
E. ☐ None of these

Space for Rough Work

41. Which of the following statements is incorrect?

Statement 1: Producers are heterotrophs.

Statement 2: Dead flesh eating animals are called carnivores.

Statement 3: The hippopotamus is known as the river horse.

Statement 4: Non-green plants are autotrophs.

A. ☐ Statements 1 and 4 only
B. ☐ Statements 1, 2 and 3 only
C. ☐ Statements 1 and 2 only
D. ☐ Statements 1, 2 and 4 only
E. ☐ None of these

42. Match the following:

	Column A		Column B
i.	*Collar bone*	*a.*	*Ulna*
ii.	*Forearm bone*	*b.*	*Femur*
iii.	*Thigh bone*	*c.*	*Clavicle*
iv.	*Lower leg bone*	*d.*	*Tibia*

A. ☐ i → d, ii → c, iii → a, iv → b
B. ☐ i → c, ii → a, iii → b, iv → d
C. ☐ i → b, ii → c, iii → d, iv → a
D. ☐ i → c, ii → d, iii → b, iv → a
E. ☐ None of these

43. Read the following sentences:

i. They have long roots that penetrate deep inside the soil in search of water.

ii. Their leaves are flat, large and stomata are found on upper surfaces.

iii. Their stems are long, narrow and have big air spaces.

iv. Their leaves are reduced to spines.

Choose the pair of sentences that are correct for the adaptations found in desert plants.

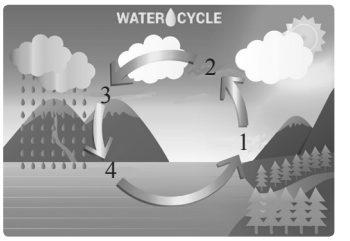
A. ☐ i and ii only
B. ☐ ii and iii only
C. ☐ i and iv only
D. ☐ ii, iii and iv
E. ☐ None of these

44. Fill in the blanks with the correct combination of words.

__a__ were discovered by an old __b__ named __c__. He had a __d__ stick. The stick had an iron casing at its lower end which was attracted by a rock. This rock which behaved like a magnet was given the name __e__.

A. ☐ a→Magnetite, b→Cobbler, c→Magnet, d→Plastic, e→Magnes
B. ☐ a→Magnes, b→Painter, c→Magnet, d→Rubber, e→Magnetite
C. ☐ a→Magnets, b→Shephered, c→Magnes, d→Wooden, e→Magnetite
D. ☐ a→Magnets, b→Cobbler, c→Magnes, d→Wooden, e→Magnetite
E. ☐ None of these

45. Arrange the processes involved in the water cycle in correct sequence on the basis of the figure given below.



A. ☐ Evaporation, Condensation, Precipitation, Accumulation
B. ☐ Condensation, Evaporation, Precipitation, Accumulation
C. ☐ Evaporation, Condensation, Accumulation, Precipitation
D. ☐ Precipitation, Evaporation, Condensation, Accumulation
E. ☐ None of these

46. Which one of the following statements is correct about scavengers?

A. ☐ Scavengers are the dead animals which do not depend on plants or other animals for their food.
B. ☐ Scavengers are the animals which hunt other animals for their food.
C. ☐ Scavengers are the animals which prepare their food by photosynthesis.
D. ☐ Scavengers are the animals that feeds on dead organisms.
E. ☐ None of these

47. Which one of the following statements is correct?

Statement I: The electric circuit provides a complete path for electricity to pass between the two terminals of an electric cell.

Statement II: When the filament of a bulb gets broken, the bulb is said to be fused.

Statement III: The electric circuit in which there is a gap in the connections between the terminals of a cell, wires and bulb is called an open circuit.

Statement IV: Those materials which allow electric current to pass through them are called insulators.

A. ☐ Statements I and II only
B. ☐ Statements I and III only
C. ☐ Statements I, II and IV only
D. ☐ Statements I, II and III only
E. ☐ None of these

48. Which one of the following statements is incorrect about the parts of a flower?

I. The stamen is the female part of a flower.

II. The pistil is the male part of a flower.

III. Pistil is known as the 'carpel'.

IV. The pollen grains contain male sex cells of a plant.

A. ☐ I, II and III only B. ☐ I and II only

C. ☐ I and IV only D. ☐ I, II, III and IV

E. ☐ None of these

49. Fill in the blanks.

Plants having leaves with reticulate venation have __a__ roots whereas those having leaves with parallel venation have__b__roots. The wheat plant has__c__venation and the sunflower plant has__d__venation.

A. ☐ a. tap, b. fibrous, c. parallel, d. reticulate

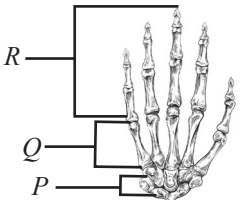
B. ☐ a. fibrous, b. tap, c. parallel, d. reticulate

C. ☐ a. tap, b. fibrous, c. reticulate, d. parallel

D. ☐ a. fibrous, b. tap, c. reticulate, d. parallel

E. ☐ None of these

50. Look at the following figure and choose the correct option.



A. ☐ P → phalanges, Q → carpals, R → metacarpals

B. ☐ P → metacarpals, Q → carpals, R → phallanges

C. ☐ P → carpals, Q → metacarpals, R → phallanges

D. ☐ P → carpals, Q → phallanges, R → metacarpals

E. ☐ None of these

Space for Rough Work

ANSWERS
INTERNATIONAL OLYMPIAD OF SCIENCE 2019 (LEVEL 2) - CLASS 6

1.	(A)	2.	(A)	3.	(B)	4.	(C)	5.	(D)	6.	(B)	7.	(B)	8.	(C)	9.	(C)	10.	(D)
11.	(A)	12.	(A)	13.	(B)	14.	(B)	15.	(C)	16.	(B)	17.	(E)	18.	(C)	19.	(A)	20.	(B)
21.	(B)	22.	(C)	23.	(A)	24.	(B)	25.	(C)	26.	(B)	27.	(B)	28.	(C)	29.	(A)	30.	(D)
31.	(B)	32.	(D)	33.	(A)	34.	(B)	35.	(A)	36.	(A)	37.	(B)	38.	(B)	39.	(A)	40.	(B)
41.	(D)	42.	(B)	43.	(C)	44.	(C)	45.	(A)	46.	(D)	47.	(D)	48.	(B)	49.	(A)	50.	(C)